

Assignment B3: Kalman Filter

Important: Before you start, read the instructions on Canvas!**1 Are extra sensors redundant?****[65 marks]**

Suppose we have a dynamical system which is described by

$$x_{t+1} = Ax_t + w_t, \tag{1a}$$

$$y_t = Cx_t + v_t, \tag{1b}$$

where x_t is the system state, w_t is a process noise, y_t is the measured output, and v_t is the measurement noise. The standard assumptions of the Kalman filter hold, that is,

1. $\mathbb{E}[w_t] = 0$, and $\mathbb{E}[v_t] = 0$ for all $t \in \mathbb{N}$
2. x_0 , $(w_t)_t$ and $(v_t)_t$ are mutually uncorrelated random variables
3. w_t is normally distributed with $\text{Var}[w_t] = Q$
4. v_t is normally distributed with $\text{Var}[v_t] = R$
5. x_0 is normally distributed with $\mathbb{E}[x_0] = \tilde{x}_0$ and $\text{Var}[x_0] = P_0$

For the sake of simplicity suppose that we have only one sensor ($y_t \in \mathbb{R}$).

In terms of quality of estimation, is there any benefit in using *another* sensor of the same type¹? What about using N sensors of the same type? Provide a theoretical answer and support it with some simulations (come up with a convincing example).

State all of your assumptions and explain all steps in your answer.

¹that is, all sensors are unbiased and have the same variance

2 Kalman filter and control system

[35 marks]

Consider the following dynamical system

$$x_{t+1} = Ax_t + Bu_t + w_t, \quad (2a)$$

$$y_t = Cx_t + v_t, \quad (2b)$$

where x_t is the system state, u_t is the input, w_t is a process noise, y_t is the measured output, and v_t is the measurement noise. Suppose that the standard assumptions of the Kalman filter are satisfied and (A, B) is stabilisable. Let K be a stabilising gain for (A, B) , i.e., the matrix $A + BK$ is stable². The state of the system is estimated with a Kalman filter. The Kalman filter produced the estimates $\hat{x}_{t|t}$, which are used to determine the control actions

$$u_t = K\hat{x}_{t|t}. \quad (3)$$

The closed-loop system with the aforementioned Kalman filter and linear controller is illustrated in Figure 1.

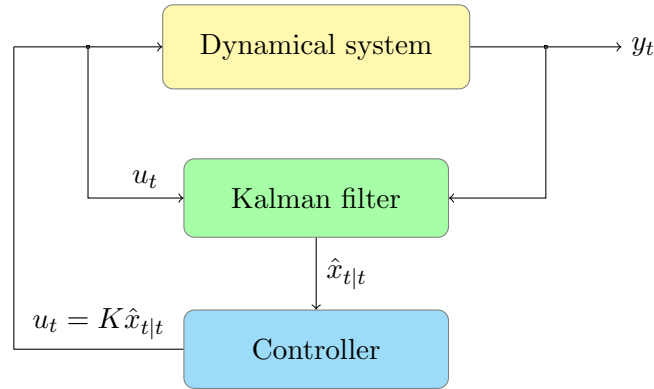


Figure 1: Closed-loop system with a Kalman filter and a controller.

(i) Start by choosing $A \in \mathbb{R}^{2 \times 2}$, $B \in \mathbb{R}^2$, $C \in \mathbb{R}^{1 \times 2}$, $Q \in \mathbb{S}_{++}^2$, $R > 0$, and a stabilising gain $K \in \mathbb{R}^{1 \times 2}$, and simulate the closed-loop system³. In your report you should include plots of y_t , $\hat{x}_{t|t}$, and u_t and you should comment on your results. In your report you should also include the equations of the Kalman filter. This first question will allow you to get an understanding of how the system works.

(ii) Is it true that the sequence $\hat{x}_{t|t}$ converges (under certain assumptions)?⁴ What about the sequences y_t and u_t ?

(iii) For a general system of the form (2) with the controller of the form (3), under what conditions do the sequences of matrices $\Sigma_{t+1|t}$ and $\Sigma_{t|t}$ converge?

²that is, all the eigenvalues of $A + BK$ are strictly within the unit circle.

³Make sure that you choose A , B and C so that (A, B) is controllable and (A, C) is stabilisable.

⁴Based on your simulations, does it converge in your simulations in Question (i)?